Lab 1

Measurement, Uncertainty, and Temperature

Say my name. Say my name.

Equipment List

- Leslie Cube
- Glass Thermometer
- IR Thermometer "gun" (2)
- Stopwatch
- Ruler
- Computer

Overview

In this weekly lab you will primarly work in pairs, with your "partner" at the same table, but feel free to discuss things with the "group" of four at your adjoining table. We will also share data with the entire class, usually using google surveys and sheets. You will turn in your own individual lab assignment, with copies of all data and graphs, in a single pdf document uploaded to the Blackboard "assignment" for that week.

1.1 Introduction and Definitions

Today we will conduct our first measurements of physical phenomena, mostly temperature measurements, and understand the necessity of the uncertainty and units associated with those measurements.

> A **physical measurement** contains three equally important parts: the numerical **value**, the **uncertainty** (also referred to as *error*) in that value, and the **units** in which these are measured.

Some examples are:

- The human body temperature is 37.0 ± 0.5 °C.
  Both the value and uncertainty are in "degrees celsius". The uncertainty value, $0.5°C$*, implies an uncertainty range about the measurement. Here, the uncertainty range is* $[36.5°C, 37.5°C]$.
- A college basketball game has 40 minutes ± 0.1 seconds of playing time.
  The final minute clock measures to tenths of a second. Note that units can be different between the value and uncertainty (but must be the same type of quantity, e.g., time).
- The Sun is 93 million miles away
  Here the uncertainty is expressed implicitly as "plus or minus a few of the last digit given", so plus or minus a few million miles.

Let's make a precise definition of the uncertainty/error in a measurement:

> Given a particular measurement, the **uncertainty** is a statement of the range of values in which another scientist is likely to make the same measurement.

Sometimes the best way to estimate the uncertainty is to actually consider a set many repeated measurements. The *standard deviation* from statistics is one technical way to express uncertainty: the "plus or minus value" that includes 68% of measurement values about their average. Without being so technical, if we can view a set of measurement values by eye (e.g., spread out on a number line), we can: (a) estimate the center of those values as the average (the reported measurement value), and (b) choose a "plus or minus value" about that average such that the uncertainty range includes most, but not all of, the measurements.

There are a number of **sources of uncertainty**:

- **Instrumental Uncertainty**: A measurement device generally has a smallest unit division shown. For an analog device — like a ruler with millimeter marks, or a thermometer with degrees — we can attempt to make a measurement *between* the smallest markings and then state an instrumental uncertainty that agrees with our assumed accuracy in that attempt (e.g., 0.3 millimeters on a ruler, or 0.3°C on a thermometer). For a digital device, we can say that the instrumental uncertainty is "plus or minus a few of the last digit" (e.g., ± 0.03 seconds on a standard stopwatch).
- **(Manufacturing) Instrumental Uncertainty**: Sometimes although an instrument shows a precise demarcation, it is clear from looking at multiple "identical" instruments that they were not manufactured (or calibrated) to that precision, e.g., multiple identical thermometers read slightly different values. The range of manufacturing uncertainty can be assessed by using multiple copies of the same device.
- **Methodological Uncertainty**: Sometimes our precision is limited more by the way we make a measurement than by the instrument itself. For example, although a stopwatch has instrumental uncertainty in the hundreths of a second, our human reaction time is no better than a few *tenths* of a second, so our methodological uncertainty would be ± 0.3 s. This can be assessed by making repeated measurements.
- **Natural Variation** Almost always, the object of measurement does not have an infinitely precise value (e.g., the "temperature outside right now" depends on where exactly you measure it, and at which moment in time; the "diameter of a rock" that is not spherical). This uncertainty can sometimes be limited by a more precise specification of the measurement (e.g., "the temperature in the shade, outside Padnos, at 10:40 am"; or "the largest

diameter of the rock"). And, like the methodological uncertainty, it can be assessed by repeated measurements.

1.2 Experiments

1.2.1 The time of my name

Have your partner recite their full name to you at a steady relaxed pace. Measure the amount of time this takes using the stopwatch. Try this multiple times and record your data below. Then switch roles and recite your own name for them.

Write down the name, and report your measurement of the recitation time with uncertainty and units.

Briefly explain how you arrived at the uncertainty value, and make mention of all "sources" of uncertainty you think might be important.

1.2.2 "Seeing" the uncertainty I: Room Temperature

One table has been set up with an array of thermometers. Since they are all localized in the same place, they should be making approximately the same measurement: the temperature of the air at that table right now.

- Take a careful clear photo of the array of thermometers, then return to your desk and write down each of the temperature measurements on a sheet of paper.
- Considering the range of measurements you have, use a ruler to put labeled tick marks on the axis below such that the values will be spread out across the line. [Looking at your maximum and minumum values, choose convenient upper and lower values on the axis and then subdivide.] At the far right, label the axis (e.g., "Temperature (°C)").
- Place a dot on/near the axis for every measurement of temperature. You can "scatter" these a bit above/below the axis for clarity.

State your final measurement of temperature, including value, uncertainty, and units. Estimate the uncertainty using your graphical display of points.

Choose one thermometer and state its *instrumental* uncertainty. Explain.

What do you think is the primary source of the uncertainty? Mention each type of uncertainty source.

1.2.3 "Seeing" the uncertainty II: Body Temperature

Here each of you will make a measurement of your body temperature using the Infrared (IR) thermometer "gun" on the palm of your hand. Note the picture on the side of the IR thermometer gun describing its "beam" of measurement (imagine a spot being projected onto the thing you are measuring, and make sure the spot is contained within the object). You will produce a two-picture plot, using a provided code (linked within the Labs section of Blackboard) and the Google CoLab implementation of python programming language.

What is the instrumental uncertainty associated to the IR thermometer?

Do you observe any natural variation of temperature across your hand?

Write down three measurements here (with value, uncertainty, and units): palm of your hand, back of your hand, and cheek/forehead. Then make three submissions to the google survey (linked within the Labs section of Blackboard).

Computational Plotting Procedure:

- Download the data from the whole class (posted on Blackboard once everyone has submitted) as a csv file. Using Excel or google sheets (`sheets.google.com`), open this file and copy only the temperature values data into the first column of a new spreadsheet (you can omit the header/title). Clean/edit this single data column (it is possible that some of the submitted data entried have typos or non-numerical characters). Save as a new csv file.
- In a web browser, go to `colab.research.google.com` and open a "New Notebook". You should be logged into your GVSU Google account.
- Paste in the code provided on blackboard for *Scatter and Histogram Plot*. If there are multiple code blocks, use the "`Code +`" button to open a new block for entry.
- To run the code, either press the "play" button on each block, or press shift-enter/shift-return. It will then prompt you for a single-column csv file to upload. Once you upload the file, the code will produce a plot on the screen.
- Notice that there are generic axis labels and titles. Examine the code and find where the axis text has been entered. Edit that text in the code to make the axis labels and titles specific to your particular data. Then re-run the code to get a new plot.

- To save and print your plot, you can either right-click and save to a file or screenshot to a file. Print out this plot so that you can add it to your scanned submission.

On the scatter plot you produced, label the approximate average value with a star, and draw two bars around it (like outstretched arms) to indicate the uncertainty range.

What are the values of the average and uncertainty? How does this uncertainty compare to the one you estimated for your own measurement(s)?

1.2.4 Leslie Cube

Procedure:

- Fill up the Leslie cube with hot/warm water at the sink and then replace the rubber stopper. Fill a small beaker with cold water.
- Place the glass thermometer first in the cold water, and then insert it into the Leslie cube such that its bulb is in the middle of the cube.

Record the glass thermometer measurements (values, uncertainties, units) of the temperature in the cold water and in the Leslie cube.

Describe how quickly the thermometer came to equilibrium with the cold water and with the hot water.

Take and record temperature measurements of <u>each side</u> of the Leslie cube using the IR thermometer gun.

1.2.5 Temperatures of other objects

Procedure:

- Using the IR thermometer gun, record measurements of temperature (below, with value, uncertainty, units for each) for the following objects. Each student (not just one per pair) should make their own measurements.
  - The surface of your table.
  - Something else in the room.
  - Outside: Something that might be in equilibrium with the air temperature.
  - Outside: Something that is probably not in equilibrium with the air temperature.
  - Outside: The sky (if you can, try to get both clear and clouded regions).
- Report the requested values to the second google survey posted on Blackboard.
- Choose one of the temperature data sets in the full-class data (see Blackboard once everyone has completed the survey) and make another Scatter-Histogram plot, like you did for the body temperature data. Adjust axis labels and titles if necessary. Save and print your resulting plot to be included with your lab submission.

Record your own temperature measurements here, and give any necessary explanations or notes of anything interesting.

1.3 Summary Questions

Brief answers are fine, but please write in complete sentences.

1. State the temperature measurements you made for the Leslie cube, including each type of side and the direct glass thermometer measurement. Speculate on why any of the values were significnatly different values, if you find that to be the case.

2. Looking at either the hand-temperature data or the other plot you made with the entire class data: how did your own estimated uncertainty (for your measured value) compare to the "spread" of the data taken by all students? Should these agree? Why or why not?

3. Two students make a measurement of the distance from the Bookstore to the Clock Tower. They are reported as follows

$$199.65 \text{ m} \pm 0.3 \text{ m}$$
$$202 \text{ m} \pm 3 \text{ m}$$

(i) On the axis below (next page), draw equally-spaced tick marks over a wide enough range to show both measurements *and their ranges of uncertainty*. (ii) Plot the two measurements using a star for the "value" and two horizontal bars (outstretched arms) to show the uncertainty. (iii) Are these two measurements "the same" or different, taking into account the uncertainty of each? Explain.

4. Shown below is a summary of the Global Average Temperature over most of human recorded history. The uncertainties for each data point are indicated by bars. Speculate on how these uncertainties might have been determined, how they change over time, and why.